

SCOPE

Scalable Chemistry Optimized through Plasma Electrification

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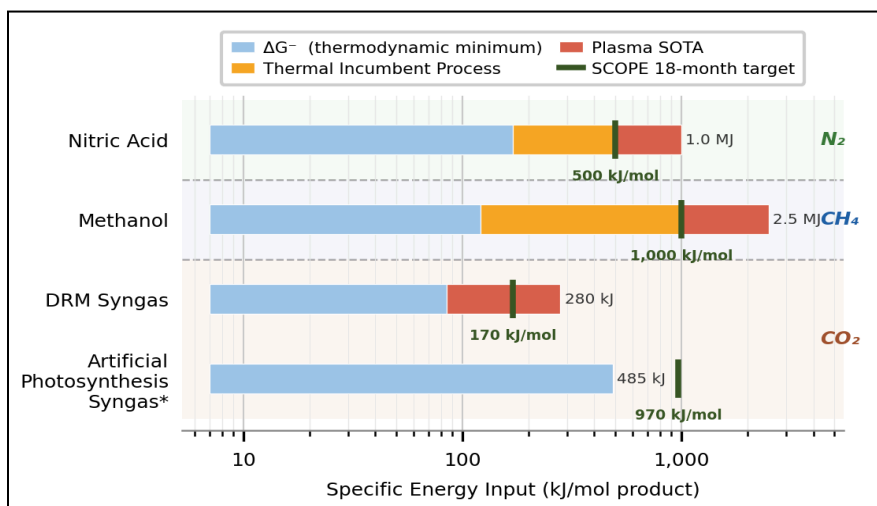
ExhiBiTS Concept Note¹

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Industrial chemical manufacturing is ~8% of US primary energy and ~5% of US greenhouse-gas emissions, but no single product dominates the footprint: ammonia, methanol, ethylene, propylene, and chlorine each contribute ~10%. What unifies them is not the product but the *intermediate*, molecular hydrogen. 95% of the world's 100 Mt/yr H₂ is fossil-derived from steam methane reforming (SMR) at 8–10 kg-CO₂ per kg-H₂ (~1 Gt/yr CO₂e), making H₂ the largest single source of process CO₂ in the sector. Because H₂ or syngas is fungible, the SMR operates at 1000t/day it increases the scale of every downstream operation (e.g. a new HNO₃ plant produces 3,000 t/day at \$17B overnight CapEx). Electrification is touted as a method to decarbonize, however today's electrification options preserve this centralized, H₂-routed architecture and carry a primary-energy penalty due to the spark-spread (the difference in cost between 1 unit of gas heat and 1 unit of electrical heat).

SCOPE is a 3-year, \$40M ARPA styled program that converts abundant atmospheric and stranded feedstocks (N₂, CO₂, CH₄) directly into valuable commodity chemicals (nitric acid, methanol, oxygenates, syngas, acetylene) **without ever generating H₂ as a feedstock or intermediate**. Plasma reactors will be co-designed with advanced power electronics (SiC & GaN) and guided by AI accelerated surrogate models. These new reactors and plasmas will deposit electrical energy surgically into the vibrational modes of these molecules, **routing chemistry through pathways forbidden in thermal equilibrium**. Example chemistries and the SCOPE targets are provided in figure 1 and the appendix table.



H1. What are you trying to do?

SCOPE will replace the incumbent hydrogen-dominated, 10+ billion dollar, 1000ton/day scale infrastructure to make today's commodity chemicals with small, electricity-only reactors that pull nitrogen from the air and convert CO₂ and natural gas directly at smaller scales more efficiently at point of use making fertilizer, methanol, oxygenates, syngas, and olefins on-site at tens of kilograms per day. Two structural changes drive this program: **(1) removing hydrogen as a chemical intermediate**, the single largest source of process CO₂ in the chemical sector, and **(2) shrinking the unit of production** from a 3,000 t/day plant to a 10 kW reactor that fits behind a renewable/point source installation. **This program will target a portfolio of chemicals using nitrogen, methane, and CO₂ chemistry** to maximize decarbonization impact. Together these eliminate the dominant emissions source of the industry and break the centralization that makes it expensive to electrify, difficult to decarbonize, and impossible to deploy on stranded feedstocks (such as wellheads and landfills) or demand centers (such as farms).

¹ Developed through the [BiTS](#) program, a part of [Renaissance Philanthropy](#).

H2. How is it done today, and what are the limits of current practice?

Industrial chemistry is built on H_2 (or syngas: $CO + H_2$) from a Steam Methane Reformer (SMR) as a universal building-block intermediate. Ammonia (Haber–Bosch), methanol (SMR $\rightarrow$ syngas $\rightarrow$ ICI process), liquid hydrocarbons (Fischer–Tropsch), and nitric acid (Haber–Bosch $\rightarrow$ Ostwald) all route through H_2 before being reacted toward the desired product. This fungibility creates two consequences that together make decarbonization extraordinarily difficult.

SMR derived H_2 is the carbon problem. It is the largest single source of process CO_2 in the chemical sector [1]; 95% of the world's 100 Mt/yr H_2 is fossil-derived from SMR at 8–10 kg- CO_2 per kg- H_2 (~1 Gt/yr CO_2 e). Chemical manufacturing is ~8% of US energy demand and ~5% of US greenhouse-gas emissions, split 70/30 between fuel and process emissions [1]. **No single product dominates** therefore any meaningful electrification and decarbonization approach must address a portfolio of chemistries.

SMR derived H_2 is the scale problem. SMR's economies-of-scale (>1,000 kg/day) increases the scale of every downstream operation. A new HNO_3 plant produces ~1,000-3,000 t/day at \$10's of billions overnight CapEx, and methanol, ammonia are similarly sized. Hazardous-material transport of ammonia or nitric acid adds up to 20% to the final product price. These plants exist where the gas is, not where the fertilizer is needed.

Today's electrification options are stuck behind this same H_2 wall due to the "spark-spread"²: the price gap between fossil heat and electric heat. Electrified Haber–Bosch carries a substantial primary-energy penalty over the SMR-fed incumbent even at curtailed-renewable prices [2], and electrically heated SMR eliminates fuel emissions but keeps the centralized, process- CO_2 -emitting architecture intact. Aqueous electrochemistry and biology cannot activate the strongest bonds ($N\equiv N$ at 945 kJ/mol; $C-H$ in CH_4 at 439 kJ/mol; $C=O$ in CO_2 at 805 kJ/mol) at industrial rates. **Methane is uniquely difficult**: any partially oxidized intermediate has weaker $C-H$ bonds than CH_4 itself, so the system runs to CO_2 unless the product is removed faster than it re-reacts. Sulfur poisons catalytic, aqueous, and biological systems alike — 0.5% of US methane is flared because Permian gas at 200+ ppm H_2S is uneconomic to send to centralized H_2 .

H3. What is new in your approach and why do you think it will be successful?

Plasmas can deliver electrical energy directly into the vibrational and electronic modes of N_2 , CO_2 , and CH_4 , lowering effective activation barriers and **routing chemistry through pathways forbidden in thermal equilibrium** [3]. Tuning the reduced electric field allows >80% of input electron energy to deposit into vibrational excitation of N_2 or CO_2 [4], ideally suited to bond cleavage by quasi-resonant vibrational climb, **the lowest physical activation energy for these stable molecules**. SCOPE makes foundational commodity chemicals from captured/stranded CO_2 , CH_4 , and air **without ever generating molecular H_2** or syngas directly attacking both the carbon problem and the scale problem of thermochemistry.

Plasma physics, power electronics, and AI capabilities make now the time to invest. The field today operates plasma reactors as black boxes brute force tuning voltage, frequency, gas composition even when the chemistry demands precision targeting of reactive states. The 2021 NASEM Plasma Science Decadal Study identified application-driven low-temperature plasma physics as a national whitespace [5]. SCOPE answers it with four research wedges run jointly by every team: **(i) state-resolved diagnostics** measuring vibrational and electronic populations *in operando*; **(ii) impedance matching** at sub-microsecond timescales; **(iii) vibrational-state engineering** via tailored sub-breakdown waveforms (the recent ns-pulse/RF "hybrid" discharge achieved >2 $\times$ improvement in NH_3 generation [6]); and **(iv) AI-accelerated kinetic surrogates** validated against (i)–(iii). Together these turn the plasma into a designed system rather than an empirically tuned arc.

Power electronics are the enabling technology that has reached sufficient maturity. SiC and GaN power switches at <\$1/W [7] switching at 100 kHz–10 MHz make spiker-sustainer and short-pulse waveforms cheap enough to co-design with reactors and make CapX/kg comparable to incumbent technologies. The modular nature allows **number-up rather than size-up** reactors which is the direct antidote to H_2 -driven economies of scale. AI surrogates accelerate design-of-experiment loops by orders of magnitude [8].

² Additional figures are available upon request

Recent results demonstrate possibilities: 1.8 MJ/mol-N for $\text{N}_2 \rightarrow \text{NO}_x$ at 94% NO_2 selectivity [9]; one-step $\text{CH}_4 + \text{H}_2\text{O} \rightarrow \text{CH}_3\text{OH}$ at room temperature [10] with 96.8% liquid-phase selectivity at plasma–catalyst–liquid interfaces [11]; 50–60% liquid oxygenate selectivity from $\text{CO}_2 + \text{CH}_4$ [12]; 33% energy efficiency for pulsed-microplasma dry methane reforming [13]. See figure for example portfolio chemistries of interest to SCOPE and appendix for more information.

H4. What is the impact if successful? If you are successful, what difference will it make?

Successful execution rewires the most carbon-intensive chemical industries onto distributed, electrified, fossil-feedstock-optional architectures. **Nitric acid from air:** 60–70 Mt/yr global production, 80% as fertilizer feedstock; eliminating the $\text{SMR} \rightarrow \text{Haber–Bosch} \rightarrow \text{Ostwald}$ supply chain closes the \$1B/yr US ammonia trade deficit and decentralizes production to where the farms that consume fertilizer are. **Decentralized methanol and oxygenates without syngas:** US methanol consumption is 12 Mt/yr; direct $\text{CH}_4 \rightarrow \text{CH}_3\text{OH}$ eliminates the SMR step, while $\text{CO}_2 + \text{CH}_4 \rightarrow \text{oxygenates}$ valorizes captured CO_2 and runs on biogas at landfills and farms where pipelines are uneconomic **turning two methane waste streams into feedstock**. Second-order benefit across all three: plasma reactors ramp from cold start to full power in seconds, suiting them to behind-the-meter intermittent renewables where curtailment-driven ramp is the operating norm.

SCOPE's 18-month targets is to demonstrate plasma energy efficiencies that are competitive at \$0.08 / kWh (US industrial average) and natural gas at <\$4/MMBTU³. The difference between the price of electricity and natural gas is known as the “spark spread” and as this gap diminishes (with more renewables and electrification) plasma chemistry becomes more appealing. **The combined target footprint can address 4% of US annual greenhouse-gas emissions and ~5% of US primary energy use.**

H5. What are the risks?

Energy yield, selectivity, and balance-of-plant must be optimized simultaneously and these tradeoffs are uncharacterized for plasmas. The program assumes electricity will be both decarbonized *and* competitive in price with fossil energy all while near-term datacenter demand is pushing prices up. The **number-up vs. size-up question:** at what scale does plasma's CapEx-light architecture lose its advantage to a centralized incumbent remains unanswered and is the single largest commercialization risk. Plasma reactors must also demonstrate industrial-qualification lifetimes (10^3 – 10^4 h) against electrode erosion and dielectric aging. Power-supply cost trajectories are favorable but not guaranteed. Cross-section data for many C/H/O/N reactions is sparse which is being addressed directly by Technical Area 2.

H6: How much will it cost? H7: How long will it take? H8: What tests signal success?

A 3-year, \$40M program with two Technical Areas funds 8–10 project teams and 2 capabilities teams.

Constraints: Hydrogen and/or syngas are disallowed feedstocks. Operating pressure >0.1bar.

Specific Energy Input (SEI): 18months: 1kW/kg feed, **36-month target** is 10kW/kg feed.

Technical Area 1 (Reactor Design): Each team includes a plasma chemist, a power electronics designer, and a modeler.

Technical Area 2 (Capabilities Team, ~\$7M, 2 teams): swarm-experiment cross-section measurements [15] for unknown C/H/O/N reactions, public reaction-rate database stewardship in the spirit of LXCat, and a shared diagnostics team to convert any project's reactor from a black box into a state-resolved measurement. Electricity is priced at \$0.05/kWh (curtailed renewable).

Program success is measured per chemistry against either the thermo-catalytic incumbent (for endo-exothermic reactions such as HNO_3) or the theoretical minimum (for endothermic reactions such as DRM or acetylene). Per-chemistry targets are given in the table below [16, 17, 18]. Other chemistries in the spirit of the program are in-scope with aggressive targets.

³ This is the equivalent of \$0.001 / kWh.

Primary Feedstock	Chemistry	ΔG Limiting step (kJ/mol)	Thermal Incumbent (MJ/mol)	Plasma SOTA (MJ/mol)	SCOPE 18-mo Target (MJ/mol) @ 1kW	SCOPE 36-mo Target (MJ/mol) @ 10kW
N ₂	HNO ₃ (N ₂ +O ₂ +H ₂ O)	170 *	0.50 (SMR-HB-Ostwald) [2]	0.42 [9]	<0.50; NO _x yield & conc. thresholds required	<0.50 + BoP;
CH ₄	C ₂ H ₂ (2CH ₄)	311	1.03 (Sachsse-Bartholome) [4]	0.94 (Hüls arc) [21]	<1.03; at ≥1 kW	<1.03 + high C ₂ H ₂ selectivity
	CH ₃ OH (CH ₄ +H ₂ O)	121	1.0 (SMR-MeOH) [6]	2.5 [10]	<1.0	<1.0 + BoP
CO ₂	DRM Syngas - (CO ₂ +CH ₄)	85 [8]	N/A [9]	~0.28 [20]	2×ΔG = 0.170	<0.170 + 99% conversion
	Artificial Photosynthesis Syngas - CO ₂ +H ₂ O	485 [23]	N/A	TBD [19]	2×ΔG = 0.970	0.970

Table 1. SCOPE program metrics by feedstock category. Energy values are per mol of primary target product. "Syngas pair" = 1 mol CO + 1 mol H₂. BoP = balance of plant. ΔG values at 298 K, 1 atm. *170 kJ/mol reflects the endothermic rate-limiting step (N₂→2NO). Overall rxn ΔG° ≈ +38 kJ/mol per HNO₃ is deceptively low due to exothermic downstream steps.

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